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Deep Learning and Applications

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Compressed Representation

- Sequential processing process data step by step.
  1. Learning temporal dependencies across the sequence.
  2. Each time step corresponds to one element in the input sequence.
  3. The model updates its hidden state to remember past information.
  4. Struggle to retain information across long sequences.

Capturing the global structure and semantic essence.

- Compressed Representation
  1. Learn a compressed latent representation of the entire input.
  2. Capturing the **global structure** and **semantic essence**.

Compressed Representation

- **Supervised learning:** using labeled data,

$$\mathcal{D}_{\text{sup}} = \{(x^{(r)}, y^{(r)})\}_{r=1}^{R}$$

learn a mapping $f_\theta : x \rightarrow y$ that minimizes prediction error.

- **Unsupervised learning:** using unlabeled data,

$$\mathcal{D}_{\text{unsup}} = \{x^{(u)}\}_{u=1}^{U}$$

discover hidden patterns or representations in x without labels.

Reconstruction:

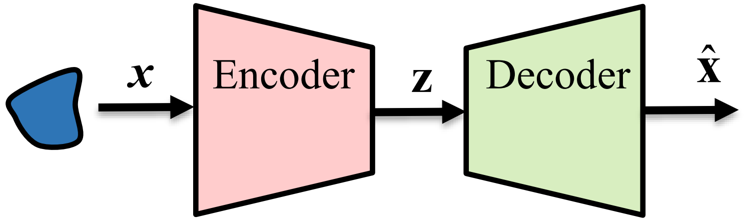
Transmitting or storing information efficiently.

$$x \xrightarrow{\text{Encoder}} z \xrightarrow{\text{Decoder}} \hat{x}$$

Autoencoder (AE)

- Objective: make reconstruction $\hat{x}$ as close as possible to x.
- The Autoencoder **learns a compact encoding** of the data:
  1. **Encoder** extracts meaningful latent features z
  2. **Decoder** learns to **rebuild the full data** from those features.

$$x \xrightarrow{\text{Encoder}} z \xrightarrow{\text{Decoder}} \hat{x}$$



- Loss function
  1. MSE: $L_{\mathrm{AE}} = ||x - \hat{x}||^2$ (Measures pixel-wise / value-wise difference)
  2. Cross-Entropy Loss:

$$L_{\mathrm{AE}} = -\sum_i [x_i \log \hat{x}_i + (1 - x_i)\log(1 - \hat{x}_i)]$$

- Dimensionality Reduction (PCA, but nonlinear)

  Learns to represent high-dimensional data with fewer variables.
- Feature Learning:

  Automatically discovers useful patterns and features.

Autoencoder (AE)

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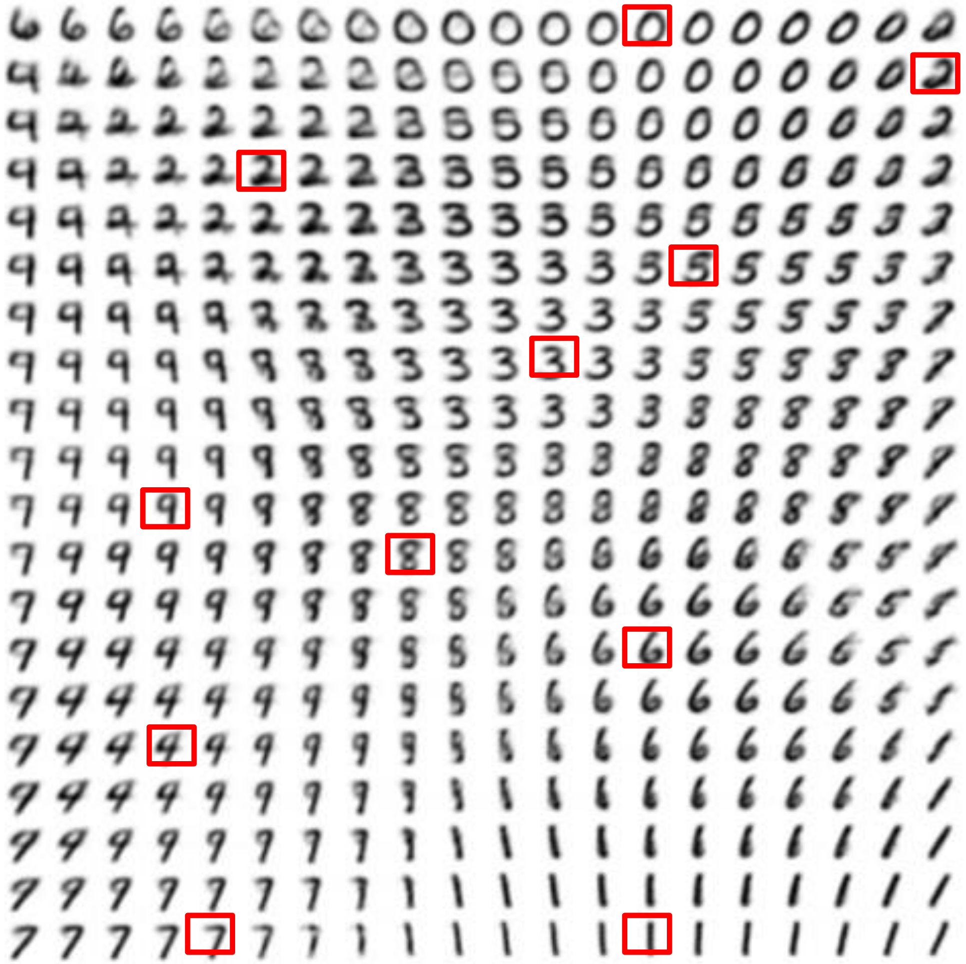
- Limitations
  1. Unstructured Latent Space
     a. The latent vector *z* has no meaningful organization or distribution.
     b. The model does not learn a true latent probability distribution.
  2. Not Generative
     a. Sampling a random *z* usually produces meaningless outputs.
     b. The decoder cannot reliably generate new valid data samples.
  3. High Risk of Overfitting
     a. The model may simply **memorize** the input data instead of learning the underlying data distribution.

Latent Distribution

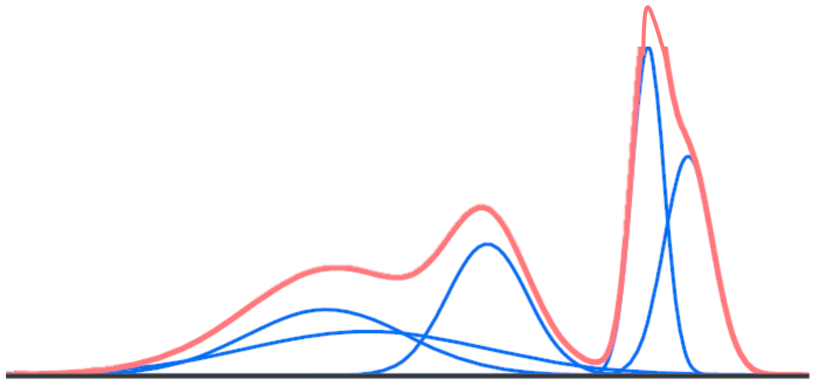
- Deterministic model
  1. AE only learns to **compress each input into a single vector**
     — it does **not** learn the structure of the latent space.
  2. The latent space may be sparse, discontinuous, or fragmented
     — only regions near training samples produce valid outputs.
  3. The latent space consists of **isolated points**

  → interpolation is possible mathematically, but **not guaranteed to be meaningful**.



(a) The two-dimensional map of the MNIST manifold

- Generative model
  1. The latent space is constrained to follow a **continuous probability distribution**
  2. All samples are forced to **occupy a shared, smooth latent space**
     → dense, continuous, and topologically organized.
  3. Interpolation now corresponds to moving **along a valid data manifold**



Variational Autoencoder (VAE)

- The latent variable z is a continuous generative space → learn a **posterior distribution over** ***z***

$$p_\phi(z \mid x) = \frac{p_\phi(x \mid z)\, p(z)}{p_\phi(x)}$$

intractable

This distribution captures both the **shape of the latent space** and the **uncertainty associated with each possible value of** ***z***.

- Assume a distribution

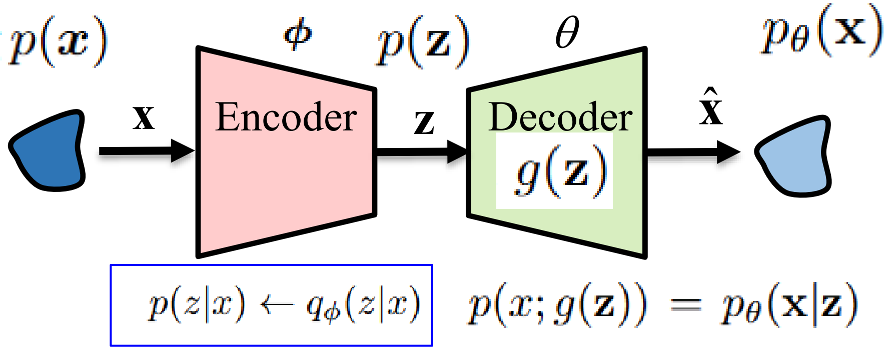
$$q_\phi(z \mid x) \approx p_\phi(z \mid x)$$

as close as possible to the true posterior.

This distribution captures both the **shape of the latent space** and the **uncertainty associated with each possible value of** ***z***.

Sample and then decode:

$$z \sim q_\phi(z \mid x), \qquad \hat{x} \sim p_\theta(x \mid z)$$



x	Observed data (e.g., images, speech, text)
z	Latent variable
$p(x)$	True data distribution (unknown)
$p(z)$	Prior distribution
$p(z \mid x)$	True posterior (intractable)
$q_\phi(z \mid x)$	Encoder approximate posterior
$p_\theta(x \mid z)$	Decoder likelihood / generative distribution
ϕ	Encoder parameters (weights)
θ	Decoder parameters (weights)
$g(z)$	Decoder neural network mapping

Variational Autoencoder (VAE)

- A posterior distribution over z

$$\log p(\boldsymbol{x}) = \log p(\boldsymbol{x}) \int q_\phi(\boldsymbol{z} \mid \boldsymbol{x})\, d\boldsymbol{z}$$

$$= \int q_\phi(\boldsymbol{z} \mid \boldsymbol{x})(\log p(\boldsymbol{x}))\, d\boldsymbol{z}$$

Rewrite as an expectation over q_ϕ

$$= \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}[\log p(\boldsymbol{x})]$$

Based on the joint distribution

$$= \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{p(\boldsymbol{z} \mid \boldsymbol{x})}\right] = \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z}) q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}{p(\boldsymbol{z} \mid \boldsymbol{x}) q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\right]$$

$$= \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\right] + \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\left[\log \frac{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}{p(\boldsymbol{z} \mid \boldsymbol{x})}\right]$$

Evidence lower bound L_b + KL Divergence

$$= \mathbb{E}_{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\right] + D_{\mathrm{KL}}\left(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \| p(\boldsymbol{z} \mid \boldsymbol{x})\right)$$

- **Expectation (mean)**: some function w.r.t. $\mathbf{x}$ is the "average" value that f takes on: $E_{x\sim P} = \int p(x)f(x)dx = \mu_{f(\mathbf{x})}$.

$$p(z \mid x) = \frac{p(x, z)}{p(x)} \quad \Rightarrow \quad p(x) = \frac{p(x, z)}{p(z \mid x)}$$

where the Kullback-Leibler (KL) Divergence is used to measure the difference between two probability distributions, and the value of the KL divergence is always greater than or equal to zero. Maximizing L_b is equivalent to maximizing $\log p(\boldsymbol{x})$.

Variational Autoencoder (VAE)

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- **Evidence Lower Bound (ELBO)** L_b

$$\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)q_\phi(z \mid x)}{p(z \mid x)q_\phi(z \mid x)}\right]$$

$$=\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z \mid x)}\right]+\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z \mid x)}{p(z \mid x)}\right]$$

KL divergence between the approximate posterior and the true posterior.

$$=\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z \mid x)}\right]+D_{\mathrm{KL}}\left(q_\phi(z \mid x) \| p(z \mid x)\right)$$

joint distribution

$$\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z \mid x)}\right]=\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x \mid z)p(z)}{q_\phi(z \mid x)}\right]$$

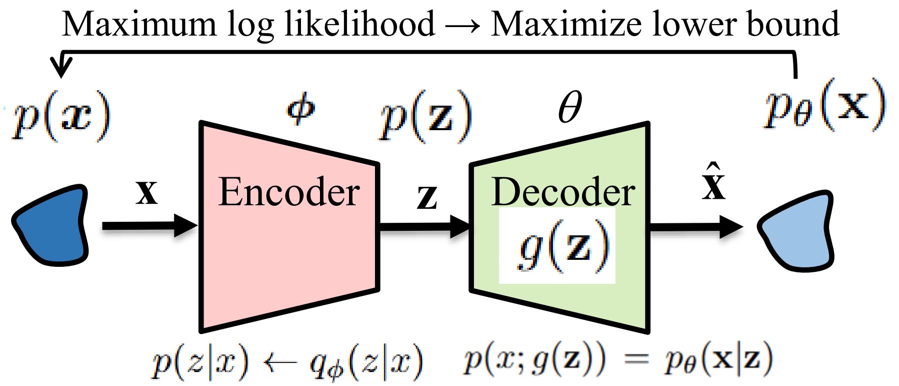
$$=\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x \mid z)\right]+\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(z)}{q_\phi(z \mid x)}\right]$$

$$=\underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x \mid z)\right]}_{\text{reconstruction term}}-\underbrace{D_{\mathrm{KL}}\left(q_\phi(z \mid x) \| p(z)\right)}_{\text{prior matching term}}$$

Variational Autoencoder (VAE)

- **Evidence Lower Bound (ELBO)** L_b

$$
\begin{aligned}
&\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)q_\phi(z\mid x)}{p(z\mid x)q_\phi(z\mid x)}\right]\\
=&\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z\mid x)}\right]+\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z\mid x)}{p(z\mid x)}\right]\\
=&\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z\mid x)}\right]+D_{\mathrm{KL}}\left(q_\phi(z\mid x)\|p(z\mid x)\right)\\
=&\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(x,z)}{q_\phi(z\mid x)}\right]=\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x\mid z)p(z)}{q_\phi(z\mid x)}\right]\\
=&\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x\mid z)\right]+\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p(z)}{q_\phi(z\mid x)}\right]\\
=&\underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x\mid z)\right]}_{\text{reconstruction term}}-\underbrace{D_{\mathrm{KL}}\left(q_\phi(z\mid x)\|p(z)\right)}_{\text{prior matching term}}
\end{aligned}
$$



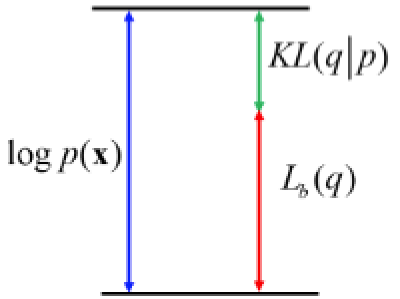
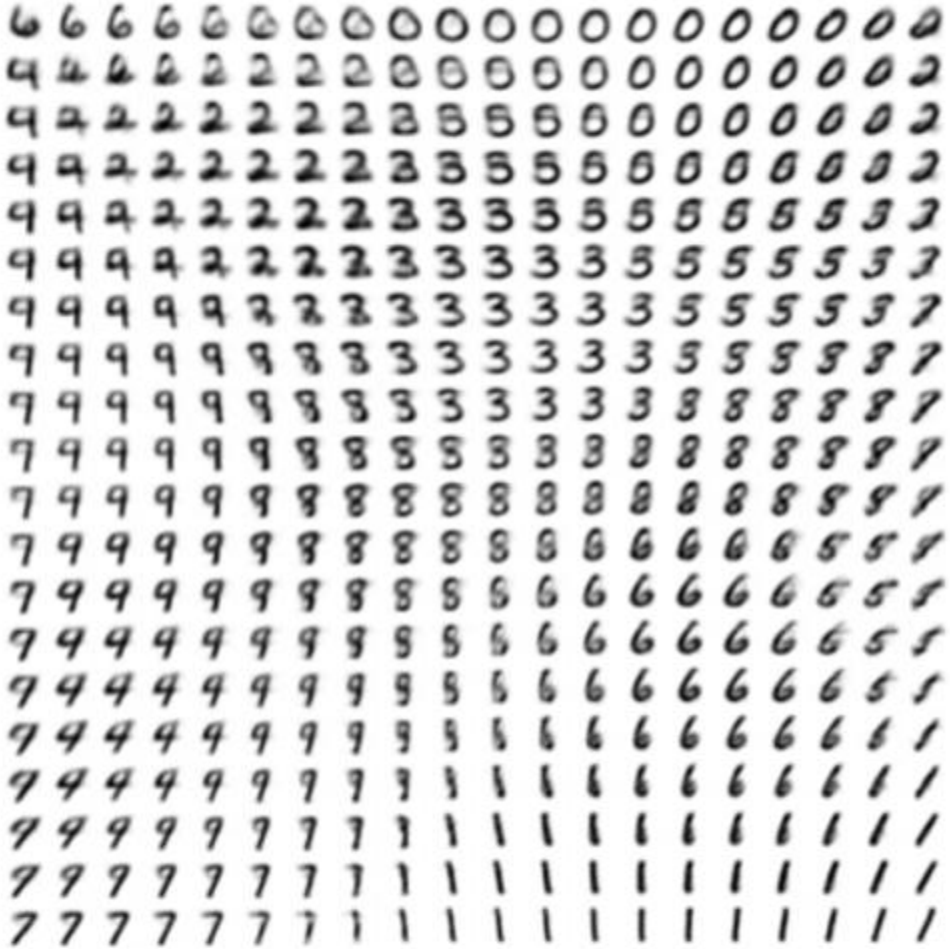
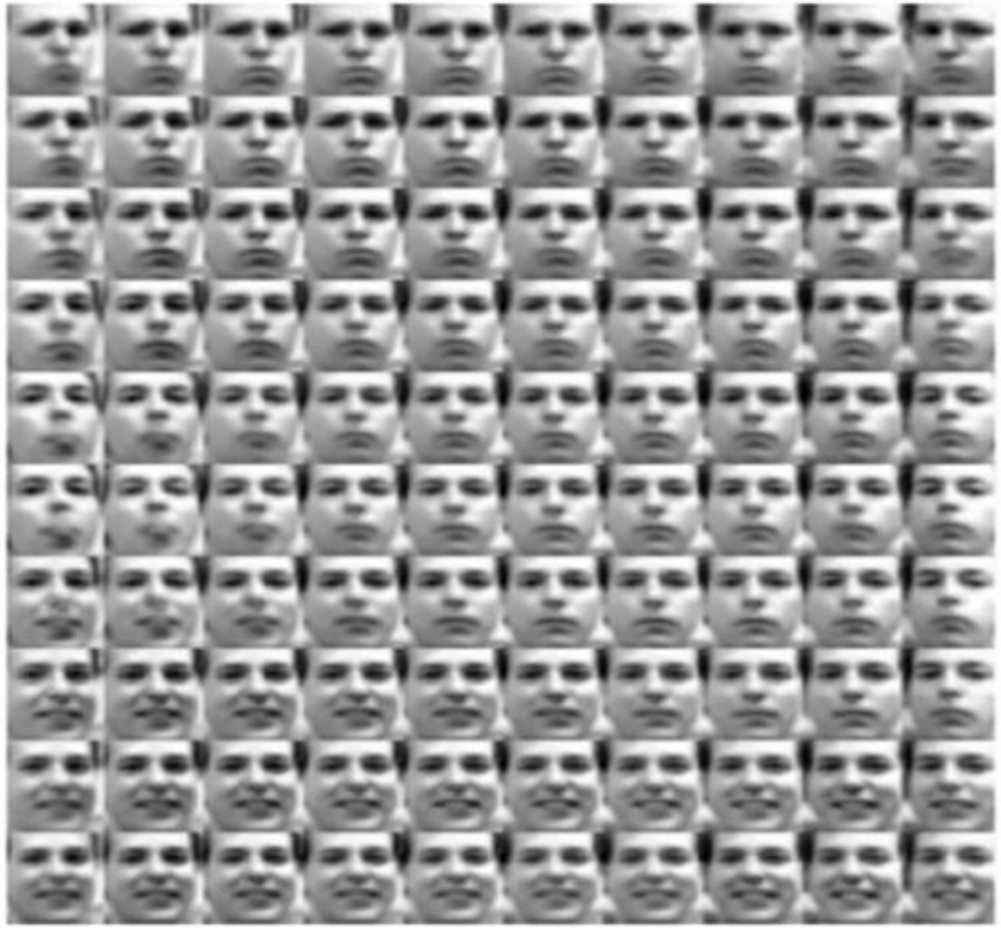


Fig. 18: The distribution $q(z)$ is held fixed, and the lower bound L_b is maximized with respect to the parameter vector ϕ, yielding a revised value ϕ_{new}. Since the KL divergence is nonnegative, this process causes the log-likelihood $\log p(x|\phi)$ to increase by at least as much as the lower bound does.

Variational Autoencoder (VAE)



(a) The two-dimensional map of the MNIST manifold



(b) The two-dimensional map of the Frey faces manifold

Fig. 19: An understanding of how the model works by training a model with a 2-D latent code, even if we believe the intrinsic dimensionality of the data manifold is much higher. The images shown are not examples from the training set but images actually generated by the model $p(x|z)$, simply by changing the 2-D "code" z (each image corresponds to a different choice of "code" z on a 2-D uniform grid).